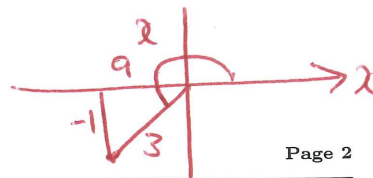


MEMO:

$$a^2 + (-1)^2 = 9$$

$$\Rightarrow a^2 = 9 - 1 \Rightarrow a = -\sqrt{8}$$



SECTION A [Multiple choice questions] - 18 marks

QUESTION 1

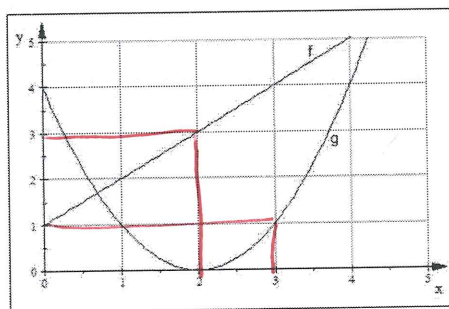
$$\cot x = \frac{1}{\tan x} = \frac{1}{\frac{-1}{\sqrt{8}}} = \frac{-\sqrt{8}}{-1} = \sqrt{8}$$

If $\sin x = -\frac{1}{3}$, $\pi < x < \frac{3\pi}{2}$, then $\cot x =$

(a) $\frac{1}{\sqrt{8}}$	(b) $-\frac{1}{\sqrt{8}}$	(c) $\frac{3}{\sqrt{8}}$	(d) $-\frac{3}{\sqrt{8}}$	☒ (e) $\sqrt{8}$	(f) $-\sqrt{8}$	(g) None of (a) - (f)
--------------------------	---------------------------	--------------------------	---------------------------	---	-----------------	-----------------------

QUESTION 2

Consider the graphs of a straight line f and a parabola g below.



$$\begin{aligned} (g \circ f)(2) &= g(f(2)) \\ &= g(3) \\ &= 1 \end{aligned}$$

$(g \circ f)(2) =$

(a) 0	☒ (b) 1	(c) 2	(d) 3	(e) 4	(f) 5	(g) None of (a) - (f)
-------	--	-------	-------	-------	-------	-----------------------

QUESTION 3

$$e^{x+\ln 3} = \frac{3}{2} \Rightarrow \ln(e^{x+\ln 3}) = \ln \frac{3}{2}$$

$$\Rightarrow x + \ln 3 = \ln 3 - \ln 2 \Rightarrow x = -\ln 2$$

If $2e^{x+\ln 3} = 3$, then $x =$

☒ (a) $\ln \frac{1}{2}$	(b) $\ln 2$	(c) $\ln \frac{1}{3}$	(d) $\ln 3$	(e) $\ln \frac{2}{3}$	(f) $\ln \frac{3}{2}$	(g) None of (a) - (f)
--	-------------	-----------------------	-------------	-----------------------	-----------------------	-----------------------

$$\ln \frac{1}{2} = -\ln 2$$

QUESTION 4

If $f(x) = \ln 2 - \ln x$, then $f^{-1}(x) =$

(a) e^{2-x}	(b) e^{x-2}	(c) $e^{\frac{2}{x}}$	(d) $e^{\frac{x}{2}}$	☒ (e) $\frac{2}{e^x}$	(f) $\frac{e^x}{2}$	(g) None of (a) - (f)
---------------	---------------	-----------------------	-----------------------	--	---------------------	-----------------------

$$y = \ln 2 - \ln x$$

$$\text{Let } x = \ln 2 - \ln y \Rightarrow \ln y = \ln 2 - x \Rightarrow y = e^{\ln 2 - x} = e^{\ln 2} \cdot e^{-x} = 2 \cdot e^{-x}$$

QUESTION 5

For which value(s) of x is $y = \cos^{-1}(1 + 2x)$ defined?

(a) $x \in [-1, 1]$	(b) $x \in [-\frac{1}{2}, \frac{\pi}{2}]$	☒ (c) $x \in [-1, 0]$	(d) $x \in [0, 1]$	(e) $x \in [0, \pi]$	(f) $x \in \mathbb{R}$
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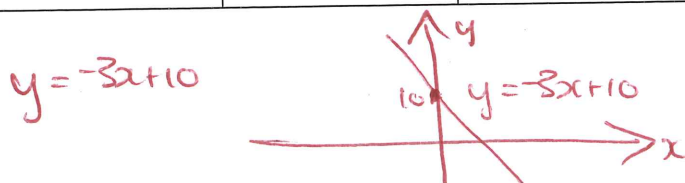
$$-1 \leq 1 + 2x \leq 1$$

$$\Rightarrow -2 \leq 2x \leq 0$$

QUESTION 6

The relationship between x and y is given by $y + 3x = 10$. What interval on the y -axis corresponds to the interval $x \in [-20, 50]$ on the x -axis?

(a) $y \in [-140, -70]$	☒ (b) $y \in [-140, 70]$	(c) $y \in [-70, 140]$	(d) $y \in [-50, 160]$	(e) None of (a) - (d)
-------------------------	---	------------------------	------------------------	-----------------------



$$x = -20: y = -3(-20) + 10 = 70$$

$$x = 50: y = -3(50) + 10 = -140$$

$$\Rightarrow y \in [-140, 70]$$

QUESTION 7

Consider the following statements:

(1) $\cos \frac{3\pi}{4} + \sin \frac{10\pi}{3} = -\cos \frac{\pi}{4} + \sin \frac{4\pi}{3}$.

(2) $\cos \frac{3\pi}{4} + \sin \frac{10\pi}{3} = -\cos \frac{\pi}{4} + \sin \frac{\pi}{3}$.

(3) $\cos \frac{\pi}{4} + \sin \frac{\pi}{4} = \sqrt{2}$.

Choose the correct answer:

- (a) Only statement (1) is correct.
- (b) Only statement (2) is correct.
- (c) Only statement (3) is correct.
- (d) Only statements (1) and (2) are correct.
- ☒ (e) Only statements (1) and (3) are correct.
- (f) Only statements (2) and (3) are correct.
- (g) Statements (1), (2) and (3) are correct.
- (h) Statements (1), (2) and (3) are false.

Realise that:

$$\sin \frac{10\pi}{3} = \sin \left(\frac{4\pi + 6\pi}{3} \right)$$

$$= \sin \left(\frac{4\pi}{3} + 2\pi \right)$$

$$= \sin \left(\frac{4\pi}{3} \right)$$

$$= -\sin \frac{\pi}{3}.$$

And: $\cos \frac{3\pi}{4} = -\cos \frac{\pi}{4}.$

ROUGH WORKING

QUESTION 8

The domain of the function $f(x) = \frac{\ln(x-3)}{1+\ln(x+1)}$ is

- (a) $(-1, \infty)$.
 (b) $(3, \infty)$.
 (c) $x \in \mathbb{R}, x \neq -1 + \frac{1}{e}$.
 (d) $(-1 + \frac{1}{e}, \infty)$.
 (e) $(-1, -1 + \frac{1}{e}) \cup (-1 + \frac{1}{e}, 3) \cup (3, \infty)$.
 (f) None of these.

① $x-3 > 0 \Rightarrow x > 3$
and ② $x+1 > 0 \Rightarrow x > -1$
and ③ $\ln(x+1) + 1 \neq 0$
 $\Rightarrow \ln(x+1) \neq -1$
 $\Rightarrow x+1 \neq e^{-1}$
 $\Rightarrow x \neq -1 + e^{-1}$

(and means we need to take the intersection of the 3 sets).

QUESTION 9

Consider $F(x) = \sec^4(x-4)$. If $F(x) = (h \circ g \circ f)(x)$, then the functions f, g and h can be represented as follows:

- (a) $f(x) = \sec x, g(x) = x-4, h(x) = x^4$.
 (b) $f(x) = \sec x, g(x) = x^4, h(x) = x-4$.
 (c) $f(x) = x-4, g(x) = \sec x, h(x) = x^4$.
 (d) $f(x) = x-4, g(x) = x^4, h(x) = \sec x$.
 (e) $f(x) = x^4, g(x) = \sec x, h(x) = x-4$.
 (f) $f(x) = x^4, g(x) = x-4, h(x) = \sec x$.

$F(x) = (h \circ g \circ f)(x)$
 $= h(g(f(x)))$
 $= h(g(x-4)) \Rightarrow f(x) = x-4$
 $= h(\sec(x-4)) \Rightarrow g(x) = \sec x$
 $= [\sec(x-4)]^4 \Rightarrow h(x) = x^4$

QUESTION 10

Consider the following statements:

- (1) $\ln a^2 b^2 = 2 \ln a \ln b$.
 (2) $\ln e^3 = 3$.
 (3) $\ln(1+e^x) = 0$.

Choose the correct answer:

- (a) Only statement (1) is correct.
 (b) Only statement (2) is correct.
 (c) Only statement (3) is correct.
 (d) Only statements (1) and (2) are correct.
 (e) Only statements (1) and (3) are correct.
 (f) Only statements (2) and (3) are correct.
 (g) Statements (1), (2) and (3) are correct.
 (h) Statements (1), (2) and (3) are false.

$\ln a^2 b^2 = \ln(ab)^2$
 $= 2 \ln ab$

$\ln e^3 = 3 \ln e = 3$.

$\ln(1+e^x)$ cannot be simplified.

QUESTION 11

What is the smallest number that would make $792 \times n$ a perfect square?

(a) 2	(b) 10	(c) 11	(d) 20	☒ (e) 22	(f) None of (a) - (e)
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QUESTION 12

The equation of the graph obtained by shifting the graph of $y = f(x)$ to the right by three units, reflecting it about the x -axis and stretching it vertically by a factor of 2, is given by:

- (a) $-2f(x+3)$.
 (b) $2f(x+3)$.
 (c) $\frac{1}{2}f(x-3)$.
 (d) $-2f(x-3)$.
☒ (e) $-2(f(x)-3)$.
 (f) $\frac{1}{2}(f(x)-3)$.
 (g) None of these.

$f(x)$
 To the right 3 units: $f(x-3)$
 Reflect about x -axis: $-f(x-3)$
 Stretch vertically by 2: $2 \times (-f(x-3))$
 $= -2f(x-3)$

End of Section A

ROUGH WORKING

Q. 11

$$\begin{array}{r}
 792 \\
 2 \overline{) 792} \\
 \underline{2 \quad 396} \\
 2 \quad 198 \\
 \underline{2 \quad 198} \\
 3 \quad 99 \\
 \underline{3 \quad 99} \\
 3 \quad 33 \\
 \underline{3 \quad 33} \\
 11 \quad 11 \\
 \underline{11 \quad 11} \\
 1
 \end{array}$$

$$\begin{aligned}
 \Rightarrow 792 &= 2^3 \times 3^2 \times 11 \\
 &= 2^2 \times 3^2 \times 2 \times 11 \\
 &= (6)^2 \times 22
 \end{aligned}$$

$792n = 6^2 \times 22 \times n$
 and for $792n$ to be a perfect square: $n=22$.

SECTION B [Fill in questions] - 8 marks

SECTIONS B AND C MUST BE SCANNED AND UPLOADED ON CLICKUP.
Please do not scan your rough working or any empty pages.

SURNAME:	MEMO
INITIALS:	
STUDENT NUMBER:	

SECTION B: Write only your final, fully simplified answer in the block provided. Only this will be marked.

QUESTION 13 [1 mark]

Assume that $a < 0$ and $b > 0$. Describe the set in which x lies if x is less than or equal to a and greater than b .

$$\boxed{\emptyset} \quad \text{or} \quad \{ \}$$

QUESTION 14 [1 mark]

Write down the equation of the function $f(x) = Be^x$ that passes through the point $(1, 2)$. Fully simplify your answer.

$\frac{1}{2}$ mark for: $\frac{2e^2}{e}$

$$\boxed{f(x) = 2e^{x-1}}$$

$$\begin{aligned} 2 &= Be^1 \\ \Rightarrow B &= \frac{2}{e} \\ \Rightarrow f(x) &= \frac{2}{e} \cdot e^x \\ &= 2e^{x-1} \end{aligned}$$

QUESTION 15 [1 mark]

If $\frac{(e^{2x})^3 e^{3x}}{e^{4x}} = e^a$, then $a =$

$$\frac{(e^{2x})^3 e^{3x}}{e^{4x}} = \frac{e^{6x+3x}}{e^{4x}}$$

$$\boxed{5x}$$

QUESTION 16 [1 mark]

Solve the inequality $\ln(x^2) - \ln 6 > 0$. Write down your answer using interval notation.

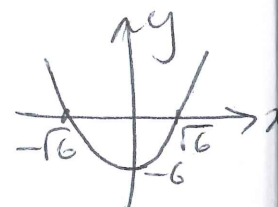
$$\boxed{x \in (-\infty, -\sqrt{6}) \cup (\sqrt{6}, \infty)}$$

$$\begin{aligned} \ln x^2 &> \ln 6 \\ \Rightarrow x^2 &> 6 \\ \Rightarrow x^2 - 6 &> 0 \end{aligned}$$

QUESTION 17 [1 mark]

Solve the inequality $\frac{|x|}{x-2} < 0$. Write down your answer using interval notation.

$$\boxed{x \in (-\infty, 0) \cup (0, 2)}$$

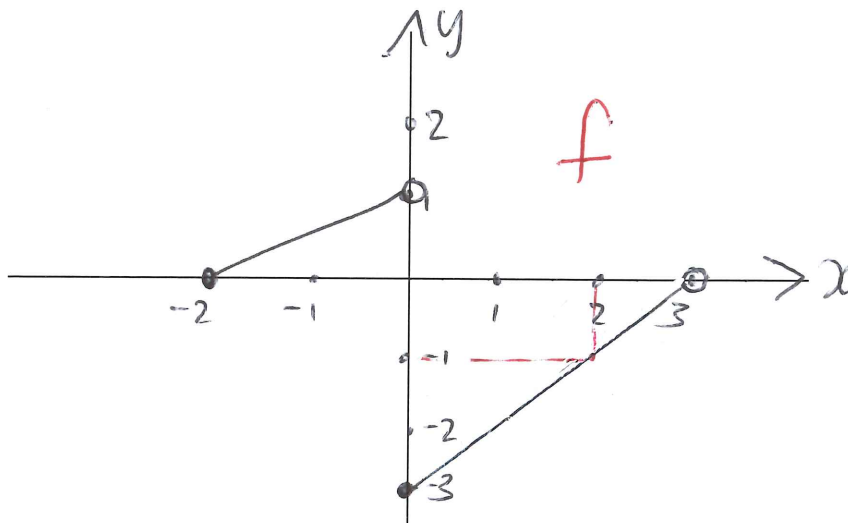


$$\begin{aligned} |x| &\neq 0 \quad \text{and} \quad x-2 < 0 \\ \Rightarrow x &\neq 0 \quad \text{and} \quad x < 2 \end{aligned}$$

QUESTION 18 [3 marks]

$$\text{Let } f(x) = \begin{cases} 1 + \frac{x}{2} & \text{if } -2 \leq x < 0, \\ x - 3 & \text{if } 0 \leq x < 3. \end{cases}$$

- (i) Sketch the graph of the function f on the axes provided, indicating all relevant intercepts and information. (1)



- (ii) Determine the domain of f^{-1} . (1)

$$\boxed{[-3, 1)} \quad \underline{\underline{or}} \quad -3 \leq x < 1.$$

- (iii) Determine $f^{-1}(-1)$. (1)

$$\boxed{2.}$$

End of Section B

ROUGH WORKING

$$\begin{aligned} \text{let } f^{-1}(-1) &= y \\ \Rightarrow f(y) &= -1 \\ \Rightarrow y &= 2 \end{aligned}$$

SECTION C [Full written answers] - 24 marks

QUESTION 19 [5 marks]

- (a) Give the definition of the absolute value of a real number
- x
- .

(1)

See p. A6 / p. 15 of the textbook.

Everything must be correct to get 1 mark.

- (b) Use the definition of the absolute value to rewrite
- $\sqrt{|x-1|+4}$
- as a piecewise defined function. Simplify completely.

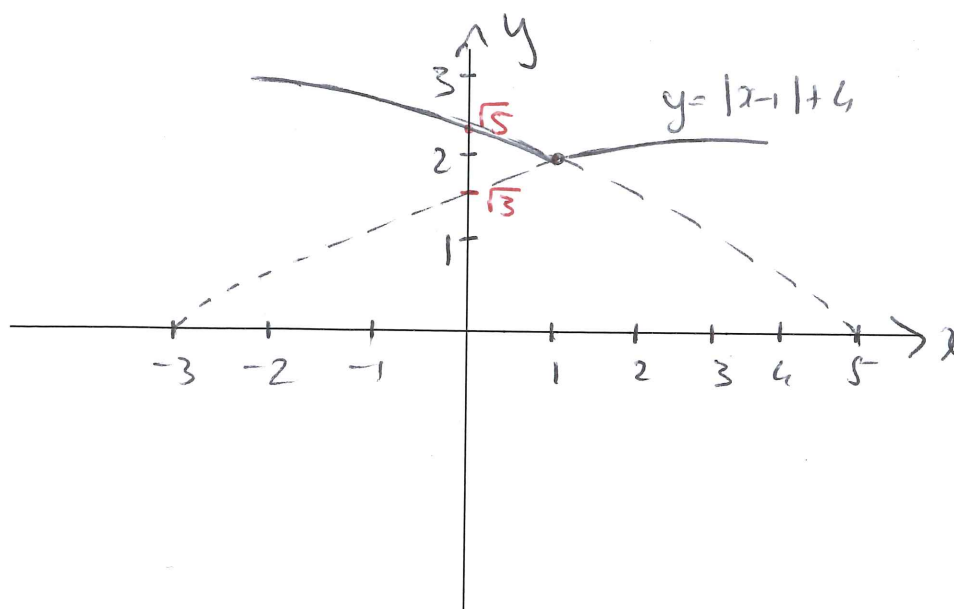
(2)

We apply the definition:

$$\begin{aligned} \sqrt{|x-1|+4} &= \begin{cases} \sqrt{x-1+4} & \text{if } x-1 \geq 0 \\ \sqrt{-(x-1)+4} & \text{if } x-1 < 0 \end{cases} \\ &= \begin{cases} \sqrt{x+3} & \text{if } x \geq 1 \\ \sqrt{-x+5} & \text{if } x < 1 \end{cases} \end{aligned}$$

- (c) Sketch the graph of the function
- $\sqrt{|x-1|+4}$
- on the axes provided, indicating all relevant intercepts and information.

(2)



QUESTION 20 [7 marks]

Solve for x . Write your answer in interval notation.

(a) $\ln(x-1) + \ln(5-x) < \ln 3$

(4)

$$\begin{aligned}
 & x-1 > 0 \text{ and } 5-x > 0 \\
 \Rightarrow & x > 1 \text{ and } 5 > x \\
 \Rightarrow & x > 1 \text{ and } x < 5 \Rightarrow \underline{x \in (1, 5)}.
 \end{aligned}$$

Also: $\ln(x-1) + \ln(5-x) < \ln 3$

$$\ln(x-1)(5-x) < \ln 3$$

$$\Rightarrow (x-1)(5-x) < 3 \text{ and } x \in (1, 5)$$

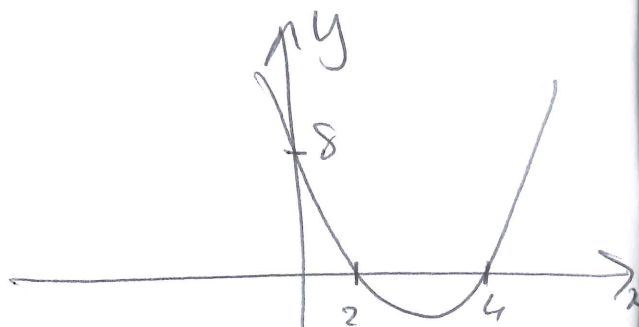
$$\Rightarrow -(x-1)(x-5) < 3$$

$$(x-1)(x-5) > -3$$

$$\Rightarrow x^2 - 5x - x + 5 + 3 > 0$$

$$\Rightarrow x^2 - 6x + 8 > 0$$

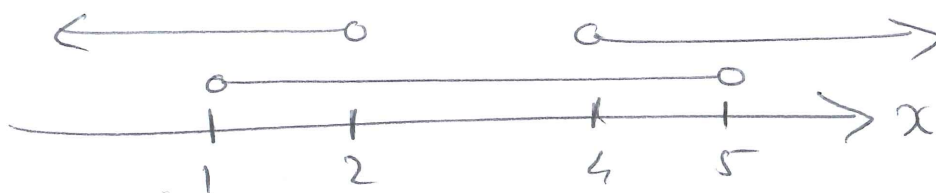
$$\Rightarrow (x-4)(x-2) > 0$$



Solution:

$$\Rightarrow x \in (-\infty, 2) \cup (4, \infty) \text{ and } \underline{x \in (1, 5)}$$

(and: intersection.)



$$\Rightarrow \underline{x \in (1, 2) \cup (4, 5)}.$$

(b) $8e^{-x} > \frac{1}{e}$

(3)

$$\begin{aligned}
 \ln 8e^{-x} &> \ln \frac{1}{e} \\
 \Rightarrow \ln 8 + \ln e^{-x} &> \ln 1 - \ln e \\
 \Rightarrow \ln 8 - x &> 0 - \ln e \\
 \Rightarrow \ln 8 - x &> -1 \\
 \Rightarrow \ln 8 + 1 &> x \\
 \Rightarrow x &< \ln 8 + 1 \\
 \Rightarrow x &\in (-\infty, \ln 8 + 1) = (-\infty, 2\ln 2 + 1).
 \end{aligned}$$

QUESTION 21 [4 marks]

(a) Complete the statement (note that $\tan^{-1} x = \arctan x$):

(1)

 $\tan^{-1} x = y$ if and only if

See p. 63

(b) Determine the value of

Everything must be correct to get 1 mark.

$\sin(2 \tan^{-1} x).$

Explain all your steps.

Similar to Example 14, p. 63.

You must use the def. of $\tan^{-1} x$ (as in (a)) here.NB to make the sketch: See Figure 24 on p. 63.where $y = \tan^{-1} x$

Final answer:

$$\begin{aligned}
 \sin(2 \tan^{-1} x) &= \sin(2y) \\
 &= 2 \sin y \cos y \\
 &= \\
 &= \\
 &= \frac{2x}{1+x^2}.
 \end{aligned}$$

QUESTION 22 [3 marks]

Consider the function

$$f(x) = \frac{x-2}{x+3}.$$

Find the inverse function, f^{-1} , together with the domain and range of f^{-1} . Use interval notation.

It is easily seen that

$$D_f = (-\infty, -3) \cup (-3, \infty)$$

$$\Rightarrow R_f = (-\infty, -3) \cup (-3, \infty).$$

Find inverse, using the method explained in 5 on p. 56.

Similar to example 4, p. 57.

Final answer: $f^{-1}(x) = \frac{2+3x}{1+x}$

and $D_{f^{-1}} = (-\infty, -1) \cup (-1, \infty).$

QUESTION 23 [3 marks]

Everything must be correct for 1 mark

- (a) Write down the definition of an odd function.

(1)

See p. 16. *f is odd if $f(-x) = -f(x)$ for all $x \in D_f$.*

- (b) Assume that
- f
- is an odd function with domain
- $\mathbb{R}$
- . Is the composite function
- $f \circ f$
- also an odd function? Show all calculations and explain your answer.

(2)

You need to start here with:

$$\begin{aligned}
 \underline{(f \circ f)(-x)} &= f(f(-x)) \\
 &= f(-f(x)), \text{ since } f \text{ is odd.} \\
 &= -f(f(x)), \text{ since } f \text{ is odd} \\
 &= -\underline{(f \circ f)(x)}
 \end{aligned}$$

 $\Rightarrow f \circ f$ is an odd function.

QUESTION 24 [2 marks]

Consider the functions

$$f(x) = a^x, a > 0, \text{ and } g(x) = e^{kx}, k \in \mathbb{R}.$$

Given that $f(x) = g(x)$, in which interval does k lie if $a > 2$.

$$\begin{aligned}
 f(x) = g(x) &\Rightarrow a^x = e^{kx} \\
 &\Rightarrow (a)^x = (e^k)^x
 \end{aligned}$$

and since the exponential function is one-to-one: $a = e^k$

$$\begin{aligned}
 \therefore \text{ If } a > 2, \\
 e^k &> 2
 \end{aligned}$$

$$\Rightarrow \ln e^k > \ln 2, \text{ because } \ln \text{ is an increasing function.}$$

End of Section C

$$\Rightarrow k > \ln 2$$

$$\Rightarrow k \in (\ln 2, \infty).$$

*You cannot
prove using
an example.*

ROUGH WORKING

Remarks.

Q23b:

- No marks are given if you try to show that this is true using an example.

Q24.

- No marks if you chose a value for a , e.g. $a=2, a=3$.
- No marks if you said: $k=\ln 2$, so $k > \ln 2$.
- You do not get marks here if your steps are not logically explained.